Avishi Agarwal

avishi.agarwal.cse@gmail.com | LinkedIn | GitHub | Portfolio

Education

Delhi Technological University | New Delhi, India — 2025 – Present
Bachelor of Technology | Computer Science Engineering — **CGPA: 9.3/10**

SHDPS | Delhi — 2023- 2025
CBSE- XII — **Class XII: 96.8%**

Salwan Public School | Gurugram — Till 2023
CBSE-X — **Class X: 97.8%**

Technical Skills

Languages: Python, C, C++, SQL
Backend & Database: FastAPI, REST APIs, PostgreSQL, SQLite, SQLAlchemy
Web & App: HTML, CSS, JavaScript, Flutter, Hugo, Streamlit
Core CS: Data Structures & Algorithms, Database Management Systems, Computer Networks, Object-Oriented Programming, Operating Systems
AI/ML: PyTorch, TensorFlow, Scikit-Learn, Graph Neural Network, RAG Pipelines, Agent Design, Embedding Models, Vector Search, Multimodal Learning, Generative AI
Core Skills & Evaluation: Model training, Model Deployment, Multi-Modal Search, Data Augmentation, Model Benchmarking, Profiling & Debugging
Tools & Systems: Git, GitHub, Linux, Docker, Kubernetes

Work Experience

Research Intern — June 2026 - Present
Delhi Technological University, New Delhi, India

- Designed and implemented PPGC, a **provenance-preserving memory system for AI agents**, evaluated across 20 conversations and 5 memory strategies, achieving **100% source preservation** while reducing stored information by 24.6% and retaining **87.3%** of relevant information.
- Validated the LLM-based evaluation framework against **57 human-labeled assessments**, achieving **94.7% agreement** and **Cohen's κ = 0.895** with human evaluators.

Academic Projects

AI NOC Copilot: Air-Gapped Predictive Copilot for Secure MPLS Operations — GitHub

- **Engineered an air-gapped AI NOC platform** over a **148-container SD-WAN/MPLS topology**, integrating SNMP, Syslog, NetFlow/IPFIX and BGP/OSPF telemetry for real-time network monitoring and predictive fault analysis.
- **Built a multi-model predictive pipeline** combining **PatchTST, STGNN+MLP, VAE+Clustering and meta-learning**, with **21 cascading fault scenarios**, supporting failure forecasting, time-to-impact estimation, shadow deployment and drift detection for safe model validation
- Implemented a **topology-aware offline RAG Copilot** combining Knowledge Graphs, incident history, runbooks and telemetry to generate **root-cause, blast-radius and remediation insights with zero outbound calls**

AASHA: Offline Rural Health Assistant (for AIC-CureBay Hackathon at APOGEE 2026, BITS Pilani) — GitHub

- Engineered an offline-first **Flutter healthcare application** integrating multilingual voice interaction, OCR and on-device AI pipelines for zero-connectivity environments.
- Developed an offline **multilingual**, voice-first LLM pipeline with quantized on-device inference in English & Hindi, alongside **OCR** for digitizing handwritten prescriptions.
- Integrated **multimodal diagnostic pathways** using audio and computer vision models for **respiratory distress, skin conditions, and anaemia screening**, enabling the system to combine multiple input modalities for broader patient assessment.

IEEE Day 2026 - Event Web Platform | React, JavaScript, Tailwind CSS, Vite, GSAP — Live Website | GitHub

- Contributed to the development and deployment of the official **IEEE Day 2026 event platform**, serving **30K+ users**, with responsive interfaces for event discovery, schedules, FAQs and registration across 9 technical events.
- Built and integrated reusable **React components and interactive UI sections**, using Tailwind CSS and GSAP-based animations while collaborating through Git/GitHub on a shared production codebase.
- Developed responsive event and team interfaces and contributed to a production website supporting IEEE student branches across **DTU and GTBIT**.

Achievements & Awards

- **Grand Finalist, Bharatiye Antriksh Hackathon (BAH) National Hackathon by ISRO:** Selected in **top 34 teams** in about **15,000+ submissions**, **Top 2** in our problem statement
- **3rd Place, Advanced Debugging (Invictus, DTU):** Identified & resolved complex logical, syntax, & runtime errors across multiple programming languages in a high-intensity competition
- **Rank 4, Campus Wide Kaggle ML Competition**: Achieved robust model generalization with minimal leaderboard variance, demonstrating strong resistance to overfitting & reliable performance on unseen test data

Positions of Responsibility

- **Executive, IEEE** — September 2025 – Present
  - Led sponsor & vendor outreach for Vihaan 9.0 & Invictus 2026; secured partnerships, including **₹1L+ deal**
  - Assisted in event planning & execution, ensuring smooth operations & team coordination
- **ML Team, AIMS** (Artificial Intelligence & Machine Learning Society, DTU) — Feb 2026 – Present
  - Working in the core ML division following a Rank among **Top 5** in a campus-wide competitive ML competition

Certifications

- **Coursera DL Specialization** by Andrew Ng — Certificate
- **Coursera ML Specialization** by Andrew Ng — Certificate